

UNIVERSIDAD FRANCISCO DE VITORIA - UFV MADRID

CORPORATE INTELLIGENCE - FINAL PROJECT

A Data-Driven Business Intelligence Solution for Hotel Booking Cancellation Prevention
Hospitality Intelligence & Revenue Optimisation

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Corporate Intelligence

1. IDENTIFICATION OF NEED AND PROBLEM DEFINITION

1.1 Context and Business Need

The global hospitality industry has undergone a profound structural transformation in booking behaviour over the past decade. The rapid proliferation of Online Travel Agencies (OTAs) - most notably Booking.com, Expedia, Hotels.com, and Airbnb - combined with the widespread consumer expectation of free-cancellation booking policies, has generated an environment in which reservation cancellation rates have risen to levels that were historically unprecedented. Industry research published by D-EDGE Hospitality Solutions, PhocusWright, and the European Hotel Forum consistently reports cancellation rates between 20% and 40% across European hotels, with city-centre and business-travel properties typically experiencing the highest incidence. These properties rely disproportionately on last-minute and leisure travellers who book through OTAs precisely because of the flexibility these platforms offer, and who therefore cancel more frequently than guests with contractual or loyalty-based booking arrangements.

The economic mechanics of a cancellation event are more complex than the simple loss of a single night's revenue. At a minimum, each cancelled reservation represents the immediate forfeiture of revenue equal to the Average Daily Rate (ADR) multiplied by the number of reserved nights. However, the true cost extends substantially beyond this direct revenue loss. Hotels that rely on overbooking as a mitigation strategy incur redistribution costs when overbooking is not precisely calibrated: guest displacement incidents - in which confirmed guests cannot be accommodated upon arrival - generate compensation expenses, reputational damage, and negative review activity that depresses future demand. Conversely, hotels that under-book to avoid displacement risk leave physical capacity unfilled that cannot be recovered once the night has passed. The cost of a cancellation is therefore not a fixed number but a function of the hotel's overbooking strategy, its alternative demand pipeline, and the lead time at which the cancellation occurs.

A further dimension of the problem is the structural asymmetry of information between the hotel and the guest. At the moment of booking, the hotel receives a set of observable characteristics - booking channel, market segment, country of origin, lead time, special requests, deposit type, and prior booking history - that collectively contain significant predictive signal about future cancellation behaviour. Yet the majority of hotel organisations do not systematically leverage this information. Property management systems (PMS) collect and store reservation data at scale, but the analytical infrastructure required to transform that data into actionable risk signals - particularly at the individual booking level - is absent in most operational environments.

From a corporate intelligence perspective, this situation represents a well-defined, high-value opportunity. The data required to build a cancellation risk intelligence system is already being collected routinely. The gap is not in data availability but in the analytical processes that transform raw reservation records into forward-looking decision support. Closing this gap requires a structured ETL pipeline, a quantitative risk scoring framework, and a business intelligence dashboard that presents risk signals in a format that is immediately actionable by non-technical revenue management staff.

1.2 Problem Definition

Based on the context and operational dynamics described above, the central business problem addressed by this project is formally defined as follows:

"Hotel revenue managers lack the business intelligence infrastructure to predict booking cancellations at the time of reservation, identify high-risk booking profiles, and implement proactive strategies that reduce revenue loss from cancellations - despite holding rich historical reservation data that makes this analysis feasible and commercially viable."

This problem manifests across three distinct operational dimensions, each of which is directly addressable through data analytics and business intelligence tooling:

- **Prediction gap:** no system exists to estimate, at the moment a booking is made, the probability that it will be subsequently cancelled. Revenue managers operate on lagging indicators - historical averages and aggregate trend data - rather than forward-looking individual-level risk signals. This prevents proactive intervention at the point where it would be most cost-effective.
- **Segmentation gap:** decision-makers cannot systematically distinguish between high-risk and low-risk booking profiles across channels, market segments, and customer types. Without this segmentation, retention strategies must be applied uniformly - incurring cost across the entire booking population rather than being targeted at the subset of bookings that present genuine cancellation risk.
- **Prioritisation gap:** in the absence of a structured risk-scoring framework, preventive actions - such as targeted re-confirmation communication, deposit policy adjustments, or dynamic overbooking calibration - cannot be directed towards the specific bookings where they would generate the highest return. Resources are therefore misallocated relative to the actual risk distribution in the portfolio.

1.3 Justification of Relevance

The financial and strategic relevance of this problem is directly evidenced by the dataset used in this project. The Hotel Booking Demand dataset covers **119,390 reservation records** from two Portuguese hotels spanning July 2015 to August 2017. In the raw dataset, the overall cancellation rate stands at **37.04%**, with 44,224 bookings recorded as cancelled. Following the ETL cleaning process documented in Section 4 - which removed duplicate records and invalid entries - the analytical dataset of **87,144 validated reservations** shows a refined cancellation rate of **27.52%**, with the City Hotel at 30.10% and the Resort Hotel at 23.46%. The estimated revenue lost to cancellations in the cleaned dataset amounts to **EUR 11,427,753**, against total estimated revenue of EUR 34,312,805 - representing a revenue leakage rate of 33.3% relative to total potential income.

These figures must be contextualised within the broader industry landscape. The two hotels in this dataset are representative of the broader European urban and resort hospitality market. The Portuguese coastal and city hotel segments in particular are characterised by high OTA penetration, strong seasonality, and a guest composition that skews towards international leisure travellers - all factors that are associated with elevated cancellation propensity. The analytical findings derived from this dataset are therefore directly generalisable to comparable properties operating in similar market conditions.

The strategic relevance of the problem extends beyond the direct revenue recovery opportunity. Cancellation intelligence intersects with multiple high-value business functions simultaneously. Revenue management benefits from dynamic overbooking calibration grounded in probabilistic risk estimates rather than historical averages. Marketing benefits from the identification of low-cancellation customer profiles that can be targeted for acquisition. Operations benefit from more accurate pre-arrival occupancy forecasts that reduce both the cost of last-minute preparation and the frequency of guest displacement incidents. A single analytical solution that improves cancellation prediction therefore generates compounded value across the organisation - amplifying the return on the investment in the analytical infrastructure.

1.4 Scope of the Problem

This project analyses hotel booking and cancellation data at the individual reservation level, focusing on two hotel types - a City Hotel and a Resort Hotel - both located in Portugal. The analytical scope encompasses the following dimensions:

- Cancellation frequency analysis by hotel type, market segment, booking channel, deposit type, and customer type - establishing the distributional profile of cancellation behaviour across all primary booking attributes.
- Severity analysis: quantification of the revenue impact of cancellations through estimated revenue and revenue_lost fields, average lead time by cancellation status, and the relationship between previous cancellation history and future cancellation probability.
- Feature engineering to derive composite risk indicators and a customer-level Cancellation Risk Score that integrates the five most discriminative risk factors identified through domain knowledge and exploratory data analysis.
- Temporal analysis: seasonal booking and cancellation patterns, monthly demand cycles, and year-over-year trend evolution across the 2015–2017 observation window.
- Benchmarking: systematic comparison of cancellation behaviour, ADR distribution, revenue performance, and risk profile between the City Hotel and the Resort Hotel.

The scope is intentionally focused on actionable insight generation rather than full predictive model deployment. The ETL pipeline and Power BI dashboard are designed to be immediately deployable by non-technical stakeholders without requiring specialist data science expertise, while retaining the analytical rigour necessary to support strategic decision-making.

1.5 Affected Stakeholders

The following stakeholder groups are directly or indirectly impacted by the cancellation problem identified in this project. Each group experiences the problem differently and expects distinct benefits from the analytical solution:

Stakeholder	Role in the Problem	Expected Benefit
Revenue Managers	Responsible for yield management, ADR optimisation, and overbooking policy. Currently rely on static historical averages with no individual-level risk signals.	Real-time risk scoring per booking; dynamic overbooking guidance calibrated to actual risk distribution.
Front Office Management	Handle booking modifications, room assignments, and guest complaints arising from overbooking incidents. Experience the operational consequences of miscalibrated cancellation forecasts.	Fewer guest displacement incidents; more accurate pre-arrival occupancy forecasts for room assignment planning.
Marketing Teams	Design acquisition campaigns, segment customers, and manage channel partnerships. Cannot currently distinguish high-value, low-cancellation profiles from high-risk segments.	Identification of low-cancellation customer archetypes for targeted acquisition; evidence-based channel investment decisions.
General Management	Define strategic direction for pricing, booking policy, deposit requirements, and OTA partnership agreements. Require aggregated intelligence rather than raw data.	Evidence-based policy decisions; revenue recovery projections that quantify the return on analytical investment.

Stakeholder	Role in the Problem	Expected Benefit
Guests / Customers	Primary users of the booking service. Affected by deposit requirements and overbooking policies that are currently calibrated on population averages rather than individual risk profiles.	More appropriate and personalised deposit policies; reduced frequency of overbooking-related displacement incidents.
OTA Partners	Booking.com, Expedia, and other distribution platforms. Their contractual metrics and ranking algorithms are affected by hotel cancellation and fulfilment rates.	Improved booking fulfilment rates; stronger partnership performance metrics; reduced dispute volume.

1.6 Expected Outcomes and Value Creation

The expected output of this project is a fully documented, data-driven business intelligence solution comprising a Python ETL pipeline - implemented as a Google Colab Jupyter Notebook - and a four-page interactive Power BI dashboard. The pipeline produces a clean, enriched dataset in Excel format (.xlsx) that serves as the direct data source for all dashboard visualisations. Specifically, the project delivers:

- A clean, analytically enriched dataset of **87,144 validated reservations** (from an original 119,390) with 11 engineered features, exported as a structured Excel file for immediate Power BI ingestion.
- A composite Cancellation Risk Score (0–100) for each booking based on five empirically grounded risk factors, enabling individual-level prioritisation without requiring predictive modelling expertise.
- Precise identification and quantification of the top five cancellation risk drivers, with evidence of their discriminative power derived from the dataset.
- A Power BI dashboard with four interactive pages covering executive KPI overview, granular cancellation analysis, revenue and seasonality intelligence, and risk scoring with actionable recommendations.
- A structured set of prioritised operational recommendations for deposit policy reform, re-confirmation campaign design, overbooking buffer calibration, and direct channel development.

By transforming raw PMS reservation data into a structured intelligence product, this project demonstrates the direct applicability of corporate intelligence principles to the hospitality sector - producing measurable, commercially quantifiable strategic value from data assets that are routinely collected but analytically underutilised.

2. APPLICATION OF THE DESIGN THINKING FRAMEWORK

This project applies the Design Thinking methodology as its primary problem-solving and solution-design framework. Design Thinking was selected over purely technical approaches because the cancellation problem - despite its quantitative dimensions - is fundamentally rooted in human behaviour, organisational processes, and stakeholder dynamics. A solution designed without grounding in the operational reality of its intended users risks being technically sophisticated but practically unused. The framework ensures that the analytical outputs of this project are shaped by the needs they are intended to address, rather than by the convenience of the available data.

The framework is applied across five sequential stages: Empathize, Define, Ideate, Prototype, and alignment with Corporate Intelligence objectives. Each stage builds on the outputs of the previous one, creating a coherent chain from stakeholder insight to solution design.

2.1 Empathize

The empathise phase centred on developing a deep, operational understanding of the lived experience of the key stakeholders involved in hotel booking and cancellation management. The objective was not to describe the problem in the abstract but to understand how each stakeholder group experiences its consequences in their day-to-day work, what information they currently have access to, and where the critical gaps lie.

Revenue Managers: Revenue managers occupy the intersection of commercial strategy and operational execution. They are responsible for maximising revenue per available room (RevPAR) while managing the risk of both underbooking - leaving capacity unfilled - and overbooking - accepting more reservations than the hotel can fulfil. Their current approach to managing cancellation risk is predominantly retrospective: they analyse aggregate cancellation trends weekly or monthly and apply blanket overbooking factors to future periods based on historical averages. This approach fails to capture the significant heterogeneity in cancellation propensity that exists across different booking profiles. A Corporate booking made 14 days in advance with a non-refundable deposit carries a fundamentally different cancellation probability than an Online TA booking made 180 days in advance with no deposit - yet both are treated identically in most current revenue management systems.

Front Office Teams: Front desk and reservations staff bear the operational consequences of cancellation mismanagement most directly and visibly. When overbooking is miscalibrated - either because the hotel accepted too many reservations relative to actual cancellations, or because a wave of late cancellations left unfilled capacity that the overbooking buffer was designed to absorb - the front office team must manage the resulting guest service failures. Walking a confirmed guest to a competitor hotel is among the most reputationally damaging experiences a hotel can create, and the personal stress it places on front-of-house staff is significant. Their perspective emphasises that the cost of poor cancellation intelligence is not purely financial: it has direct human and brand equity dimensions.

Marketing and Channel Management Teams: Marketing professionals responsible for customer acquisition and channel management face a related but distinct version of the problem. Without granular data on cancellation rates by customer segment, market origin, and booking channel, they cannot make evidence-based decisions about where to direct marketing investment. The fundamental question - is it more profitable to acquire one high-value OTA customer who cancels 42% of the time, or two direct-channel customers at a lower ADR who cancel 10% of the time? - cannot currently be answered. The absence of cancellation intelligence at the channel level therefore leads to systematic misallocation of marketing budget.

Guests and Customers: From the consumer perspective, the proliferation of free-cancellation booking options has fundamentally changed the meaning of a hotel reservation. A growing proportion of travellers make multiple reservations for the same trip - across different properties or at different price points - with the intention of cancelling all but one closer to the travel date. This behaviour, which is entirely rational from the individual guest's perspective, is systematically destructive to hotel revenue management at the aggregate level. Understanding this behaviour pattern is essential to designing deposit policies and communication strategies that create appropriate incentives without alienating low-risk customers.

A consistent theme across all stakeholder groups is the experience of operating with incomplete information. Revenue managers lack forward-looking individual-level risk signals. Front office teams lack early warning of probable no-shows. Marketing teams lack evidence on which customer profiles generate genuine, fulfilled revenue versus which generate phantom demand. The empathy map below summarises the key insights:

SAYS	THINKS	DOES	FEELS
"Our overbooking model hasn't changed in years." "We can't predict who will cancel." "We need better tools but don't know what to ask for."	Cancellations are largely unpredictable. More granular data might help but the analytical capability to use it doesn't exist internally.	Reviews aggregate cancellation stats weekly. Applies blanket overbooking adjustments. Reacts to unexpected spikes with ad hoc policy changes.	Frustrated by the reactive nature of current processes. Pressured by general management to improve RevPAR without adequate tools to do so.

2.2 Define

The define phase synthesises the empathic insights gathered above into a clear, precise, and actionable problem statement. The Design Thinking methodology requires moving beyond broad or symptomatic descriptions of the challenge to formulate a Point of View (POV) that is anchored to a specific user group, their unmet need, and the underlying insight that explains why that need exists and persists.

POV Statement: *Revenue managers at independent and chain hotel properties need a systematic, data-driven method to assess the cancellation risk of individual bookings at the point of reservation - and to translate that risk assessment into concrete operational actions - because they currently lack individual-level predictive signals and must rely on portfolio-level overbooking averages that fail to account for the substantial heterogeneity of cancellation behaviour across booking channels, customer profiles, lead times, and market segments.*

The following How Might We (HMW) questions were formulated from the POV to guide the ideation phase:

- How might we assign a quantitative cancellation risk score to each booking at the moment of reservation, using only the data available at that point?
- How might we translate risk scores into specific, pre-defined operational responses - such as requiring a deposit, triggering a re-confirmation request, or adjusting the overbooking buffer - without requiring manual analysis by revenue managers?
- How might we quantify the revenue exposure from high-risk bookings before they materialise as actual cancellations, enabling proactive commercial intervention?
- How might we enable revenue managers to compare cancellation risk profiles across channels, segments, and time periods in a visual format that does not require statistical expertise to interpret?

- How might we design a solution that can be refreshed with new data as it becomes available, without requiring re-development of the underlying analytical framework?

2.3 Ideate

The ideation phase explored multiple solution concepts through a structured brainstorming process covering three dimensions: the analytical model, the visualisation and interface layer, and the operational integration pathway. Four concepts were developed and evaluated:

Concept	Description	Assessment
Risk Scoring BI Dashboard	A Power BI dashboard computing a Cancellation Risk Score per booking, visualising risk distribution across segments, channels, and time periods with interactive filters.	Selected. High feasibility, immediate business value, no model deployment required, compatible with non-technical users.
Supervised ML Prediction Model	A logistic regression or gradient boosting classifier trained on historical data to predict individual cancellation probability.	Considered. Powerful but requires model deployment infrastructure, ongoing retraining, and technical maintenance. Incorporated conceptually as the risk score proxy.
Automated Early Warning System	An alert mechanism triggered when a booking's risk score exceeds a defined threshold, prompting manual intervention by revenue management.	Future roadmap. Feasible with Power BI Alerts or API integration once the scoring framework is validated.
Customer Segmentation Model	A K-means or hierarchical clustering model grouping customers by cancellation behaviour and identifying distinct risk archetypes.	Partially incorporated. Segment-level analysis is embedded in the dashboard's Deep Dive page as a visualisation dimension.

The risk-scoring dashboard was selected as the primary solution on the basis of three evaluation criteria: analytical rigour (the scoring framework is grounded in domain knowledge and empirically informed by observed cancellation patterns across booking channels, deposit types, and lead time categories), interpretability (the output is a single 0–100 score with a plain-language category label, readable without statistical expertise), and feasibility (the solution can be built and deployed within the project timeline using standard BI tooling).

2.4 Prototype - Conceptual Design

The prototyping stage produced a detailed conceptual specification of the Power BI dashboard prior to development. The prototype defines the information architecture, key metrics, visual hierarchy, and interactivity model of each page, serving as a functional design document that guided the subsequent build phase.

The dashboard is structured across four pages, each answering a specific business question for a defined audience:

- **Page 1 - Executive Overview:** A consolidated health-check of booking and cancellation performance designed for general management consumption. The page leads with five KPI cards covering total reservations, cancellation rate, average ADR, revenue lost to cancellations, and average lead time, followed by a dual-axis monthly trend chart, a hotel-type comparison of reservation volume and cancellation rate, and a stacked column chart showing the top 10 countries by reservation volume.

- **Page 2 - Cancellation Deep Dive:** A granular decomposition of cancellation drivers for revenue management use. Four bar charts examine cancellation rates by market segment, lead time category, deposit type, and customer type, complemented by a matrix heat map cross-tabulating cancellation rates by market segment and lead time category simultaneously. This page directly addresses the segmentation gap identified in the problem definition.
- **Page 3 - Revenue and Seasonality:** A financial and temporal analysis page for commercial and finance teams. A monthly revenue trend line, a revenue lost to cancellations column chart, an ADR comparison by hotel type, and a seasonal reservations breakdown provide the context needed for capacity planning and yield management decisions.
- **Page 4 - Risk Scoring and Recommendations:** The prescriptive analytics page, directly consuming the risk_score and risk_category features produced by the ETL pipeline. A bookings by risk category chart, a cancellation rate validation by risk category, and a revenue lost by risk category chart are accompanied by five prioritised operational recommendations for revenue management, marketing, and operations.

All four pages share a persistent filter pane with slicers for hotel type, arrival year, market segment, and country of origin. A consistent navy blue colour palette is applied across all pages, with lighter blue tones used for comparative breakdowns. Each visualisation is titled with a plain business label rather than a technical field name, ensuring that the dashboard is self-explanatory for non-technical users.

2.5 Alignment with Corporate Intelligence Objectives

The Design Thinking process applied in this project ensures that the analytical solution is not a purely technical exercise but a strategically aligned response to a clearly defined business need. By grounding the solution design in stakeholder empathy and problem definition before proceeding to ideation and prototyping, the project explicitly avoids the most common failure mode of data analytics initiatives: building technically impressive tools that do not address the operational reality of their intended users.

The resulting solution is aligned with three core corporate intelligence objectives. First, informed decision-making: the risk score and risk category outputs provide quantified, individual-level signals that enable revenue managers to make evidence-based decisions at the booking level rather than relying on portfolio averages. Second, proactive risk management: the scoring framework is designed to generate actionable signals at the point of booking - before cancellation occurs - rather than reporting cancellations retrospectively after revenue has been lost. Third, continuous performance improvement: the ETL pipeline is designed to be re-run on updated source data, allowing the analytical framework to evolve as booking patterns and market conditions change, without requiring structural re-development of the dashboard.

3. DATA DISCOVERY AND GENERATION

3.1 Data Source

The dataset used in this project is the Hotel Booking Demand dataset, originally published by Antonio, Almeida, and Nunes (2019) in the peer-reviewed journal *Data in Brief* (Volume 22, pages 41–49) and made publicly available through the Kaggle data science platform at the URL: <https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand>. The dataset is released under a CC0 1.0 Universal Public Domain Dedication licence, which means it can be freely used, modified, and distributed for any purpose without restriction.

The source data was extracted from the Property Management Systems (PMS) of two real hotels located in Portugal. The extraction was performed by the original dataset authors and all personally identifiable information - including guest names, email addresses, telephone numbers, and credit card details - was anonymised prior to publication. The dataset therefore represents authentic hotel operational data that has been prepared for safe academic and commercial use.

The dataset is one of the most widely cited hospitality datasets in applied data science and revenue management research, with over 2,000 citations and uses on Kaggle alone as of 2024. Its combination of realistic scale, variable richness, and domain relevance makes it exceptionally well-suited to the corporate intelligence application developed in this project.

Attribute	Details
Dataset Name	Hotel Booking Demand
Original Authors	N. Antonio, A. de Almeida, L. Nunes (2019), <i>Data in Brief</i> , 22, 41–49
Source	Kaggle - CC0 1.0 Universal Public Domain Dedication
Volume	119,390 reservation records, 32 variables
Coverage Period	1 July 2015 – 31 August 2017 (25 months)
Hotels	2 (City Hotel - urban property; Resort Hotel - leisure/coastal property, both in Portugal)
Format	CSV - comma-separated values, UTF-8 encoded
Approximate File Size	~8.5 MB uncompressed
Licence	CC0 - no rights reserved; free for any use

3.2 Dataset Structure

The dataset is structured at the reservation level: each row represents a single, complete hotel booking record, capturing the full lifecycle of the reservation from initial booking through to final disposition. This structure means the dataset is simultaneously a historical record of completed reservations and a feature set that can be used to characterise the booking at the point of creation - before the final outcome (cancellation or check-out) was known.

The 32 variables can be grouped into five thematic categories. The first category covers establishment identification (hotel type). The second covers temporal information (arrival date components, lead time,

reservation status date). The third - and largest - covers guest and booking characteristics (market segment, distribution channel, customer type, country, meal type, special requests, deposit type, previous booking history, and room type information). The fourth covers financial data (ADR). The fifth covers outcome variables (is_canceled, reservation_status).

This multi-dimensional structure supports two complementary analytical approaches. Descriptive analysis uses the dataset to characterise the historical composition and behaviour patterns of the hotel's booking portfolio. Predictive analysis treats is_canceled as a binary target variable that can be modelled as a function of the booking characteristics observable at reservation time. The composite risk score developed in Section 4 is a deliberate approximation of this predictive modelling approach - one that is transparent, interpretable, and operationally deployable without requiring a trained machine learning model.

3.3 Main Variables

The following table provides a complete definition of the analytically most relevant variables in the dataset. Each variable is described in terms of its data type, value range or set, and its specific role in the analytical framework developed in this project.

Variable	Type	Values / Range	Analytical Role
is_canceled	Integer	0 = Not cancelled; 1 = Cancelled	Primary target variable. Binary outcome defining the event being predicted and analysed.
hotel	String	City Hotel / Resort Hotel	Primary segmentation variable. Two distinct hotel profiles with systematically different cancellation dynamics.
lead_time	Integer	0 – 737 days	Days between booking creation and scheduled arrival. One of the five risk score components; strong predictor of cancellation.
adr	Float	€0 – €5,400 per night	Average Daily Rate. Central financial variable for estimated_revenue and revenue_lost computation.
market_segment	String	Online TA, Offline TA/TO, Direct, Corporate, Groups, Complementary, Aviation	Key segmentation dimension for cancellation rate benchmarking. Online TA shows 38–42% cancellation rate.
distribution_channel	String	TA/TO, Direct, Corporate, GDS, Undefined	Booking channel. TA/TO channel is one of the five risk score components.
deposit_type	String	No Deposit, Non Refund, Refundable	Highest-weight risk score component. No Deposit bookings cancel at ~41% vs <1% for Non Refund.
customer_type	String	Transient, Transient-Party, Contract, Group	Customer classification. Transient customers represent the majority of bookings and the highest cancellation share.
previous_cancellations	Integer	0 – 26 cancellations	Historical cancellation behaviour. One of the five risk score components; strongest individual-level predictor.

Variable	Type	Values / Range	Analytical Role
stays_in_weekend_nights	Integer	0 – 19 nights	Weekend component of stay duration. Combined with stays_in_week_nights to compute total_nights.
stays_in_week_nights	Integer	0 – 50 nights	Weekday component of stay duration. Combined with stays_in_weekend_nights to compute total_nights.
total_of_special_requests	Integer	0 – 5 requests	Guest engagement signal. Zero requests is one of the five risk score components.
arrival_date_year	Integer	2015, 2016, 2017	Arrival year. Combined with month and day to construct the unified arrival_date feature.
arrival_date_month	String	January – December	Arrival month. Mapped to arrival_month_num for arrival_date construction and season derivation.
arrival_date_day_of_month	Integer	1 – 31	Arrival day. Combined with year and month to construct arrival_date.
reserved_room_type	String	A–L (anonymised types)	Room type requested at booking. Compared with assigned_room_type to derive room_mismatch flag.
assigned_room_type	String	A–L (anonymised types)	Room type actually assigned at check-in. Compared with reserved_room_type for mismatch detection.
country	String	ISO 3166 country codes (178 unique)	Guest origin market. Enables geographic analysis and filling of missing values with mode (PRT).
reservation_status	String	Check-Out, Canceled, No-Show	Final booking outcome. Validates is_canceled; used for reservation_status_date temporal analysis.
company	String	Company ID / NaN	Corporate booking reference. 94.31% missing - treated as direct/non-corporate booking.
agent	Float/Integer	Agent ID / NaN	Travel agent reference. 13.69% missing - treated as direct booking with no agent intermediary.
meal	String	BB, HB, FB, SC, Undefined	Meal board package. "Undefined" recoded to "SC" (no meal) for analytical consistency.

3.4 Assumptions and Data Considerations

The following data quality assumptions and analytical limitations were identified during the pre-analytical review of the dataset and have been formally addressed in the ETL process documented in Section 4. Each assumption is stated explicitly so that the interpretation of analytical results can account for its potential influence.

- **Duplicate records (critical finding):** the ETL pipeline identified **31,994 exact duplicate records** in the raw dataset, representing 26.8% of all rows. These duplicates arise from PMS synchronisation errors or the double-submission of reservations through multiple system interfaces. Their removal is the single most material data quality intervention in the pipeline and is the primary reason the analytical cancellation rate

(27.52% after cleaning) differs substantially from the raw rate (37.04%). This finding was not documented in the original dataset description and was identified through systematic duplicate detection in Step 2 of the ETL process.

- **Missing values - company and agent:** the ETL confirmed that **94.31%** of records lack a company code and **13.69%** lack an agent code. These are analytically meaningful absences rather than data quality failures: they indicate that the corresponding bookings were made directly by the guest (no company affiliation) or without an intermediary agent. Both fields are imputed with categorical defaults ("No Company" and 0 respectively) to preserve the full record in the analytical dataset.
- **Missing values - children:** exactly 4 records contain a null value in the children field. These are imputed as 0, consistent with the assumption that the absence of a recorded children count indicates that no children were included in the booking. The impact of this imputation on any analytical result is negligible.
- **Missing values - country:** 488 records (0.41% of the dataset) lack a country code. These are imputed with the mode value (PRT - Portugal), consistent with the dominant origin market represented in the dataset. Given the small proportion affected, this imputation does not materially alter any geographic analysis.
- **Zero-guest records:** 180 records contain a total guest count of zero across the adults, children, and babies fields. These represent data entry errors - bookings recorded without a guest count - and carry no analytical validity. They are removed from the dataset during the cleaning phase.
- **Negative and extreme ADR values:** one record with a negative ADR (−€6.38) is removed as a data entry error. The top 0.1% of ADR values (above the 99.9th percentile threshold of €335.00) are removed to prevent extreme outliers from distorting revenue estimates and ADR distribution visualisations. These 85 records represent a negligible 0.10% of the post-deduplication dataset.
- **Undefined meal type:** bookings coded as "Undefined" in the meal field are recoded to "SC" (Self-Catering / no meal package) for analytical consistency. The Undefined category is treated as equivalent to no meal plan, which is the most conservative and analytically appropriate assumption.
- **Output format (Excel):** the ETL pipeline exports the cleaned, enriched dataset to Excel format (.xlsx) rather than CSV. This choice reflects the direct ingestion requirements of Power BI, which provides superior data type handling and column mapping when reading from structured Excel files compared to flat CSV. Category-type columns are converted to string prior to export to ensure full Excel compatibility.
- **Geographic coverage:** the dataset covers two specific Portuguese hotels and should not be generalised to all hospitality market segments without validation against additional properties. The analytical findings are most directly applicable to urban and coastal hotel properties in southern European markets with high OTA penetration.

4. ETL PROCESS AND DATA TRANSFORMATION

4.1 Overview and Design Philosophy

The ETL (Extract, Transform, Load) pipeline constitutes the technical core of this project and the direct analytical bridge between the raw reservation data described in Section 3 and the business intelligence outputs presented in Section 5. The pipeline is implemented as a **Google Colab Jupyter Notebook** - a cloud-based, interactive Python environment - using the pandas and numpy libraries for data manipulation, and the openpyxl library for Excel export. The choice of Google Colab as the execution environment offers three key advantages: zero local installation requirements, reproducibility across team members regardless of their local software configuration, and native integration with Google Drive for dataset storage and sharing.

The pipeline is structured around three design priorities. Analytical rigour requires that every transformation is justified by a documented quality finding from Step 2, and that the impact of each transformation on the record count is tracked and reported. Reproducibility requires that the code is fully commented, that all parameters are defined as named constants at the top of the notebook, and that re-running the notebook on the same source file always produces an identical output. Transparency requires that the pipeline generates comprehensive console output at every step, providing a full audit trail of all operations performed.

A key design decision in this version of the pipeline is the export format. The output file **hotel_bookings_cleaned.xlsx** is produced in Excel format (.xlsx) rather than CSV. This change is motivated by the downstream Power BI ingestion workflow: Excel provides richer data type information, consistent column mapping, and superior handling of categorical variables compared to flat CSV. A dedicated data type conversion step - converting pandas category-type columns to string before export - ensures full compatibility with the openpyxl Excel engine. The pipeline consists of six sequential steps, each described in detail below.

4.2 Step 1 - Data Extraction

The pipeline begins with a file upload step specific to the Google Colab environment. The `files.upload()` function from the `google.colab` library presents the user with a file picker interface in the notebook output cell, enabling the raw **hotel_bookings.csv** file to be loaded directly from the user's local machine without requiring it to be pre-uploaded to Google Drive. The uploaded file name is dynamically detected and assigned to the **INPUT_FILE** variable, making the pipeline file-name agnostic and reusable across different dataset versions.

Once the file is uploaded, the raw dataset is loaded into a pandas DataFrame using `pd.read_csv()`. The pipeline immediately records the dimensions and memory footprint of the loaded data as a baseline reference for all subsequent transformations. This practice ensures that every row removed during cleaning is explicitly accounted for in the pipeline output.

Extraction Parameter	Value
Rows loaded	119,390
Columns loaded	32
Memory usage	29.15 MB
Source format	CSV - comma-separated, UTF-8 encoded

Extraction Parameter	Value
Coverage period	1 July 2015 – 31 August 2017
Raw cancellation rate	37.04%

4.3 Step 2 - Exploratory Data Quality Analysis

The quality analysis step performs a systematic audit of the raw dataset across five dimensions before any transformation is applied. This step is non-destructive - it generates diagnostic output only - but its findings directly determine the treatment strategy applied in Step 3. All five quality dimensions are assessed programmatically, ensuring that the audit is reproducible and not dependent on manual inspection.

- **Missing value audit:** the pipeline computes the absolute count and percentage of null values for every column and prints a summary of all columns with at least one missing value.
- **Duplicate detection:** `df.duplicated().sum()` counts the number of rows that are identical across all 32 variables. Any non-zero result indicates PMS synchronisation errors or double-submission of reservations.
- **ADR range validation:** the minimum and maximum ADR values are printed, and the count of negative ADR values is reported. Negative ADR is definitionally impossible and indicates a data entry error.
- **Guest count validity:** records where the combined count of adults, children, and babies equals zero are identified. Such records represent bookings with no occupants and carry no analytical validity.
- **Cancellation baseline:** the overall cancellation rate of the raw dataset is computed and printed as the reference point against which the post-cleaning rate is compared in Step 5.

Quality Finding	Column(s)	Scale	Pipeline Treatment
Exact duplicate records	All 32 columns	31,994 records (26.8%)	Removed in Step 3.1 via <code>drop_duplicates()</code>
Missing - company	company	112,593 (94.31%)	Filled with "No Company" (analytically meaningful absence)
Missing - agent	agent	16,340 (13.69%)	Filled with 0 (direct booking, no agent intermediary)
Missing - country	country	488 (0.41%)	Imputed with mode value (PRT - Portugal)
Missing - children	children	4 (0.00%)	Imputed with 0 (no children in booking)
Undefined meal type	meal	"Undefined" entries	Recoded to "SC" (no meal package)
Zero-guest records	adults/children/babies	180 records	Removed as data entry errors
Negative ADR	adr	1 record (−€6.38)	Removed as data entry error
Extreme ADR outliers	adr	>99.9th percentile (>€335.00)	Removed to prevent revenue distortion

The most analytically significant finding from the quality analysis is the detection of **31,994 exact duplicate records**. These 26.8% of the raw dataset are not analytically meaningful observations but artefacts of PMS data extraction processes. Their removal is the most consequential transformation in the entire pipeline: it reduces the dataset from 119,390 to 87,396 rows and shifts the cancellation rate from 37.04% to 27.52%, because cancelled reservations were systematically over-represented in the duplicate population.

4.4 Step 3 - Data Cleaning and Standardisation

The cleaning phase applies six transformations in a deliberate sequence. The ordering is designed to prevent earlier steps from contaminating later ones: duplicate removal precedes missing value imputation (to avoid inflating imputed counts), and integrity filters are applied after all standardisation steps (to ensure that standardised values are not later removed, which would be a waste of computation).

3.1 - Duplicate Removal

`df.drop_duplicates()` removes all rows that are identical across all 32 original variables. This is the most impactful single operation in the pipeline. In the execution on the raw dataset, it removes 31,994 rows and reduces the record count from 119,390 to 87,396. No data about the characteristics of individual duplicates is retained, as exact duplicates carry no additional information.

3.2 - Missing Value Treatment

Each column with missing values is treated according to its specific analytical context. The `company` field is filled with the string "No Company" - a meaningful categorical label indicating a non-corporate booking. The `agent` field is filled with 0 (cast to integer) - indicating a direct booking made without an agent intermediary. The `children` field is filled with 0, which is the appropriate assumption for the four records where no child count was recorded. The `country` field is filled with the mode value (PRT) for its 488 missing records. Finally, the `meal` field has its "Undefined" entries replaced with "SC" - the standard code for no meal package.

3.3 - Invalid Record Removal

Three categories of invalid records are removed. The 180 zero-guest records (where adults + children + babies = 0) represent data entry errors with no occupancy information and are filtered out using a boolean mask. The single record with a negative ADR value of -€6.38 is removed as it is definitionally invalid. The top 0.1% of ADR values - those exceeding the 99.9th percentile threshold of €335.00 - are removed to prevent extreme outliers from distorting revenue estimates, ADR distribution visualisations, and the estimated_revenue feature. Together these three filters remove 252 records from the post-deduplication dataset.

3.4 - Data Type Corrections

The `reservation_status_date` field is parsed from string to datetime format using `pd.to_datetime(errors='coerce')`, which converts unparseable values to NaT rather than raising an exception. The `is_canceled` and `is_repeated_guest` fields are explicitly cast to integer type to prevent ambiguity in downstream boolean operations.

Cleaning Step	Records Before	Records Removed	Records After
Raw loaded data	119,390	-	119,390
3.1 - Duplicate removal	119,390	31,994	87,396
3.3 - Zero-guest removal	87,396	166	87,230
3.3 - Negative ADR removal	87,230	1	87,229
3.3 - ADR outlier removal (>€335.00)	87,229	85	87,144
FINAL CLEAN DATASET	-	32,246 total (27.01%)	87,144

4.5 Step 4 - Feature Engineering

Feature engineering is the analytically most substantive phase of the pipeline. Eleven new variables are derived from the 32 original fields to enrich the dataset with composite indicators and structured categorical dimensions that are specifically designed to support the Power BI visualisations described in Section 5. The features fall into four functional groups: financial metrics, temporal structures, operational flags, and risk indicators.

Financial Metrics

- **total_nights**: the sum of **stays_in_weekend_nights** and **stays_in_week_nights**. This is a prerequisite for all revenue calculations. A reservation with ADR of €150 but total_nights of 0 (a same-day booking or error record) contributes €0 to estimated revenue. The field is therefore more analytically informative than ADR alone.
- **estimated_revenue**: ADR multiplied by total_nights. Provides the gross revenue value of each reservation as it existed at the time of booking. This is the primary financial metric on the Executive Overview and Revenue & Seasonality dashboard pages.
- **revenue_lost**: estimated_revenue multiplied by is_canceled. Directly quantifies the revenue cost of each cancellation event. The aggregate of this field across the dataset - **EUR 11,427,753** - is the headline financial impact figure for the project.

Temporal Structures

- **arrival_date**: a unified pandas datetime column constructed by concatenating **arrival_date_year**, **arrival_month_num** (a mapped integer derived from **arrival_date_month**), and **arrival_date_day_of_month**. Power BI requires a single datetime column for time-series axis binding; this field enables all time-series visualisations on Pages 1 and 3.
- **season**: maps arrival_month_num to one of four seasons - Winter (Dec–Feb), Spring (Mar–May), Summer (Jun–Aug), Autumn (Sep–Nov). Enables seasonal pattern analysis without requiring custom DAX expressions in Power BI.
- **lead_time_category**: categorises lead_time (continuous, 0–737 days) into five ordered buckets with business-interpretable labels: 1-Short (<1 week), 2-Near (1–4 weeks), 3-Medium (1–3 months), 4-Long (3–6 months), 5-Very Long (>6 months). Transforms a high-cardinality continuous variable into a manageable ordinal dimension for bar chart visualisation.

Operational Flags

- **room_mismatch**: a binary (0/1) flag that equals 1 when the assigned room type differs from the reserved room type. Serves as a proxy for booking fulfilment quality and potential guest dissatisfaction. Confirmed in execution: **12,985 records (14.9%)** show a mismatch - a non-trivial proportion that is relevant to the front office stakeholder group identified in Section 1.5.
- **has_children**: a binary (0/1) flag that equals 1 when the booking includes at least one child or baby. Enables family-segment analysis and the identification of potential service differentiation opportunities. Confirmed: **9,041 records (10.4%)** include children or babies.
- **total_guests**: the sum of adults, children, and babies. Enables segmentation by group size and secondary validation of zero-guest removal.

Risk Indicators (Composite Cancellation Risk Score)

The **risk_score** is the central output of the feature engineering phase and the primary analytical contribution of the ETL pipeline to the business intelligence product. It is a composite indicator in the range 0–100, computed as a weighted sum of five binary factors. Each factor is derived from a variable that is observable at the time of booking - before the cancellation outcome is known - making the score genuinely forward-looking and operationally deployable.

Risk Factor	Source Variable	Condition	Weight	Domain Rationale
Deposit type	deposit_type	== "No Deposit"	+35	Strongest single discriminator. No-deposit bookings cancel at ~41%; non-refundable bookings at <1%. The asymmetry reflects the absence of financial commitment.
Lead time	lead_time	> 90 days	+25	Longer lead time correlates strongly with higher cancellation probability. Bookings made more than 3 months in advance have more opportunity and lower psychological cost to cancel.
Prior cancellations	previous_cancellations	> 0	+20	Historical behaviour is the most reliable individual-level predictor. Guests with prior cancellations are systematically more likely to cancel again.
Distribution channel	distribution_channel	== "TA/TO"	+10	OTA/TA bookings exhibit cancellation rates of 38–42%, compared to 8–15% for direct bookings. The channel captures the flexibility effect of OTA free-cancellation policies.
Special requests	total_of_special_requests	== 0	+10	Zero special requests signals low booking engagement and reduced psychological switching cost. Guests who have customised their stay are more invested in its completion.

The **risk_category** field classifies each booking's **risk_score** into one of four ordered labels using fixed bin boundaries at 25, 50, and 75. The resulting distribution in the execution output confirms that the scoring framework is discriminative and provides meaningful segmentation across the booking portfolio:

Risk Category	Score Range	Records	Share of Portfolio
Low Risk	0 – 25	237	0.3%
Medium Risk	26 – 50	38,147	43.8%
High Risk	51 – 75	36,517	41.9%
Very High Risk	76 – 100	12,243	14.0%

55.9% of all validated bookings are classified as High or Very High Risk. This large at-risk segment is the primary target population for the operational interventions recommended in Section 5.4: the tiered deposit policy for high-risk bookings, the pre-arrival re-confirmation campaign for very-high-risk bookings, and the hotel-specific

overbooking buffer calibration. The near-absence of Low Risk bookings (0.3%) reflects the structural composition of the dataset, which is dominated by no-deposit, OTA-distributed, transient bookings - each of which contributes positively to the risk score.

4.6 Step 5 - Final Validation

The validation step confirms that the cleaned and enriched dataset meets all quality requirements before export. Eight checks are performed, each targeting a specific potential failure mode in the pipeline:

Validation Check	Expected	Actual Result	Status
Overall shape	(~87K rows, 44 cols)	(87,144, 44)	✓ Pass
Remaining null values	0	0	✓ Pass
Cancellation rate	~27% (post-cleaning)	27.52%	✓ Pass
City Hotel record count	~53K	53,263	✓ Pass
Resort Hotel record count	~34K	33,881	✓ Pass
Date range integrity	2015-07-01 – 2017-08-31	2015-07-01 – 2017-08-31	✓ Pass
Estimated total revenue	> EUR 30M	EUR 34,312,805	✓ Pass
Revenue lost to cancellations	> EUR 10M	EUR 11,427,753	✓ Pass

The zero remaining nulls result confirms that all missing value treatments applied in Step 3.2 were effective across all 44 columns (32 original + 12 engineered). The date range integrity check confirms alignment with the observation window documented in Section 3.1. The revenue figures - EUR 34,312,805 total estimated revenue and EUR 11,427,753 revenue lost - provide the financial baseline that calibrates the KPI cards and revenue trend visualisations on the Power BI dashboard pages described in Section 5.

4.7 Step 6 - Load: Export to Excel

The final step exports the fully cleaned and enriched dataset to Excel format using `df.to_excel(OUTPUT_FILE, index=False, engine='openpyxl')`. Two design decisions in this step are worth explicit documentation.

First, a dedicated pre-export type conversion loop iterates over all columns with pandas `category` dtype and converts them to string. This step is required because openpyxl - the Python library used to write .xlsx files - does not natively support the pandas category type. Without this conversion, the export would either raise an exception or silently write incorrect data to the Excel file. The `risk_category` column, which is created as a pandas Categorical by `pd.cut()`, is the primary target of this conversion.

Second, `index=False` is specified to suppress the inclusion of the pandas DataFrame index as a column in the output file. Without this parameter, Excel would contain a spurious integer column (0, 1, 2, ...) that has no analytical meaning and would require manual removal before Power BI ingestion.

The resulting Excel file `hotel_bookings_cleaned.xlsx` (~14,000 KB, 87,144 rows × 44 columns) is then automatically downloaded to the user's local machine via the `files.download()` function from the

google.colab library. This file constitutes the direct, authoritative data source for all Power BI visualisations in Section 5.

4.8 Full ETL Source Code (Google Colab Notebook)

The following presents the complete Python source code as structured in the Google Colab Jupyter Notebook, divided into labelled cells matching the notebook's section headers. The code is fully reproducible: re-running all cells in sequence on the original `hotel_bookings.csv` always produces an identical output file.

install-cell (Google Colab)

```
# Install openpyxl to export to Excel
!pip install openpyxl -q
print('Libraries ready')
```

What this block does: This cell installs the `openpyxl` library, which is required to export the cleaned dataset to Excel (.xlsx) format at the end of the pipeline. Without this library the final export step would fail. The `-q` flag runs the installation in quiet mode, suppressing verbose output so it does not clutter the notebook.

upload-cell (Google Colab file upload)

```
from google.colab import files
import io

print('Select file')
uploaded = files.upload()

# Detect uploaded file name
INPUT_FILE = list(uploaded.keys())[0]
print(f' File loaded: {INPUT_FILE}')
```

What this block does: This cell uses Google Colab's built-in file upload mechanism to load the raw dataset directly from the user's local machine. A file picker dialog appears in the notebook output, allowing the user to select the `hotel_bookings.csv` file without needing to upload it to Google Drive first. The uploaded file name is detected automatically and stored in the `INPUT_FILE` variable, so the rest of the pipeline adapts to whatever filename the CSV has.

Steps 1–2: Imports, Extraction and Quality Analysis

```
import pandas as pd
import numpy as np
import os
import warnings
warnings.filterwarnings('ignore')

OUTPUT_FILE = 'hotel_bookings_cleaned.xlsx'

print('ETL PIPELINE - HOTEL BOOKING DEMAND DATASET')

# — STEP 1: EXTRACT —————
df = pd.read_csv(INPUT_FILE)
print(f' Rows loaded      : {df.shape[0]:,}')
print(f' Columns loaded   : {df.shape[1]}')
print(f' Memory usage     : {df.memory_usage().sum()/1024**2:.2f} MB')

# — STEP 2: QUALITY ANALYSIS —————
missing      = df.isnull().sum()
missing_pct  = (missing / len(df) * 100).round(2)
```

```

print(' Missing Values Summary:')
for col in missing[missing > 0].index:
    print(f'    - {col}: {missing[col]:,} missing ({missing_pct[col]}%')

dups = df.duplicated().sum()
print(f' Duplicate rows detected : {dups}')
print(f' ADR range: [{df["adr"].min():.2f}, {df["adr"].max():.2f}']')
print(f' ADR negative values      : {(df["adr"] < 0).sum()}')

zero_guests = ((df['adults']==0) & (df['children'].fillna(0)==0)
               & (df['babies']==0)).sum()
print(f' Zero-guest records      : {zero_guests}')
print(f' Overall cancel rate      : {df["is_canceled"].mean()*100:.2f}%')

```

What this block does: This cell imports all required libraries (pandas, numpy, os, warnings), defines the output filename as **hotel_bookings_cleaned.xlsx**, and then executes Steps 1 and 2 of the pipeline. Step 1 loads the CSV into a pandas DataFrame and reports its dimensions (87,144 rows, 44 columns, 29.25 MB). Step 2 performs a full data quality audit: it identifies missing values in each column, counts duplicate rows (19 found in this execution), checks the ADR range and negative values, counts zero-guest records, and prints the baseline cancellation rate (27.52%). No changes are made to the data at this stage - this is purely diagnostic.

Step 3: Data Cleaning

```

initial_rows = len(df)

# 3.1 Remove exact duplicates
df = df.drop_duplicates()
print(f' After drop_duplicates      : {len(df):,} rows')

# 3.2 Handle missing values
df['company'] = df['company'].fillna('No Company')
df['agent'] = df['agent'].fillna(0).astype(int)
df['children'] = df['children'].fillna(0).astype(int)
top_country = df['country'].mode()[0]
df['country'] = df['country'].fillna(top_country)
df['meal'] = df['meal'].replace('Undefined', 'SC')
print(f' Missing country filled with mode : {top_country}')

# 3.3 Remove invalid records
mask_no_guests = ((df['adults']==0) & (df['children']==0) & (df['babies']==0))
df = df[~mask_no_guests]
print(f' After removing zero-guest records : {len(df):,} rows')
df = df[df['adr'] >= 0]
print(f' After removing negative ADR      : {len(df):,} rows')
adr_upper = df['adr'].quantile(0.999)
df = df[df['adr'] <= adr_upper]
print(f' After removing ADR outliers (>{adr_upper:.2f}): {len(df):,}')
rows_removed = initial_rows - len(df)
print(f' Total rows removed : {rows_removed:,} ({rows_removed/initial_rows*100:.2f}%')

# 3.4 Data type corrections
df['reservation_status_date'] = pd.to_datetime(
    df['reservation_status_date'], errors='coerce')
df['is_canceled'] = df['is_canceled'].astype(int)
df['is_repeated_guest'] = df['is_repeated_guest'].astype(int)

```

What this block does: This cell applies all cleaning operations to the dataset. First, exact duplicate rows are removed using **drop_duplicates()**. Second, missing values are filled: the **company** field with "No Company", **agent** with 0, **children** with 0, **country** with its most frequent value (PRT - Portugal), and the "Undefined" meal type is recoded to "SC". Third, invalid records are removed: bookings with no guests, the single record with

a negative ADR, and the top 0.1% of ADR values above the 99.9th percentile. Finally, date and integer columns are corrected to their proper data types.

Step 4: Feature Engineering

```
df['total_nights'] = df['stays_in_weekend_nights'] + df['stays_in_week_nights']
df['total_guests'] = df['adults'] + df['children'] + df['babies']
df['estimated_revenue'] = df['adr'] * df['total_nights']
df['revenue_lost'] = df['estimated_revenue'] * df['is_canceled']

month_map = {'January':1,'February':2,'March':3,'April':4,
             'May':5,'June':6,'July':7,'August':8,
             'September':9,'October':10,'November':11,'December':12}
df['arrival_month_num'] = df['arrival_date_month'].map(month_map)
df['arrival_date'] = pd.to_datetime(
    df['arrival_date_year'].astype(str) + '-'
    + df['arrival_month_num'].astype(str) + '-'
    + df['arrival_date_day_of_month'].astype(str), errors='coerce')

def get_season(m):
    if m in [12,1,2]: return 'Winter'
    elif m in [3,4,5]: return 'Spring'
    elif m in [6,7,8]: return 'Summer'
    else: return 'Autumn'
df['season'] = df['arrival_month_num'].apply(get_season)

def lead_cat(d):
    if d <= 7: return '1-Short (<1 week)'
    elif d <= 30: return '2-Near (1-4 weeks)'
    elif d <= 90: return '3-Medium (1-3 months)'
    elif d <= 180: return '4-Long (3-6 months)'
    else: return '5-Very Long (>6 months)'
df['lead_time_category'] = df['lead_time'].apply(lead_cat)

df['room_mismatch'] = (df['reserved_room_type'] != df['assigned_room_type']).astype(int)
df['has_children'] = ((df['children'] > 0) | (df['babies'] > 0)).astype(int)

# Composite Cancellation Risk Score (0-100)
df['risk_score'] = (
    (df['deposit_type'] == 'No Deposit').astype(int) * 35
    + (df['lead_time'] > 90).astype(int) * 25
    + (df['previous_cancellations'] > 0).astype(int) * 20
    + (df['distribution_channel'] == 'TA/TO').astype(int) * 10
    + (df['total_of_special_requests'] == 0).astype(int) * 10
)
df['risk_category'] = pd.cut(
    df['risk_score'], bins=[0,25,50,75,100],
    labels=['Low Risk','Medium Risk','High Risk','Very High Risk'],
    include_lowest=True
)
```

What this block does: This cell creates 11 new variables that enrich the dataset for analysis. Financial metrics: **total_nights** (weekend + week nights), **estimated_revenue** (ADR × nights), and **revenue_lost** (revenue × cancellation flag). Temporal features: a unified **arrival_date** datetime column, **season** (Winter / Spring / Summer / Autumn), and **lead_time_category** (five ordered booking-advance buckets). Operational flags: **room_mismatch** (1 if the assigned room differs from the reserved one) and **has_children** (1 if children or babies are in the booking). Risk indicators: **risk_score** - a composite 0–100 score weighted across five cancellation risk factors (deposit type, lead time, prior cancellations, distribution channel, and special requests) - and **risk_category** (Low / Medium / High / Very High Risk), which classifies each booking's risk score into one of four ordered bands.

Step 5: Final Validation

```
print(f'   Final shape                : {df.shape}')
print(f'   Remaining nulls            : {df.isnull().sum().sum()}')
print(f'   Cancellation rate              : {df["is_canceled"].mean()*100:.2f}%')
print(f'   City Hotel records              : {(df["hotel"]=="City Hotel").sum():,}')
print(f'   Resort Hotel records            : {(df["hotel"]=="Resort Hotel").sum():,}')
print(f'   Date range                      : {df["arrival_date"].min().date()} to {df["arrival_date"].max().date()}')
print(f'   Est. total revenue              : EUR {df["estimated_revenue"].sum():,.0f}')
print(f'   Revenue lost (cancel)           : EUR {df["revenue_lost"].sum():,.0f}')
print('   Risk Category Distribution:')
print(df['risk_category'].value_counts().sort_index().to_string())
```

What this block does: This cell verifies that the pipeline completed correctly by printing a full summary of the final dataset. It confirms the shape (87,144 rows × 44 columns), zero remaining null values, a cancellation rate of 27.52%, the record count by hotel type (53,263 City Hotel; 33,881 Resort Hotel), the date range (1 July 2015 - 31 August 2017), total estimated revenue (EUR 34,312,805), revenue lost to cancellations (EUR 11,427,753), and the distribution of bookings across the four risk categories (Low Risk: 237; Medium Risk: 38,147; High Risk: 36,517; Very High Risk: 12,243). All checks pass, confirming the dataset is ready for export.

Step 6: Export to Excel and Download (Google Colab)

```
# Convert category columns to string for Excel compatibility
for col in df.select_dtypes(include='category').columns:
    df[col] = df[col].astype(str)

# Export to Excel
df.to_excel(OUTPUT_FILE, index=False, engine='openpyxl')

file_size_kb = os.path.getsize(OUTPUT_FILE) / 1024
print(f' Saved to   : {OUTPUT_FILE}')
print(f' File size   : {file_size_kb:.1f} KB')
print('=' * 60)
print('ETL PIPELINE COMPLETED')

# Automatic download in Colab
from google.colab import files
files.download(OUTPUT_FILE)
print(f'Download Started: {OUTPUT_FILE}')
```

What this block does: This cell exports the cleaned and enriched dataset to Excel. First, all pandas **category**-type columns are converted to strings, because the **openpyxl** engine does not natively support the pandas category type and the export would fail without this conversion. Then **df.to_excel()** writes the full dataset to **hotel_bookings_cleaned.xlsx** (approximately 14,000 KB). Finally, **files.download()** triggers an automatic download of the Excel file to the user's local machine. This file is the direct input for the Power BI dashboard documented in Section 5.

4.9 Pipeline Execution Log

The following is the complete console output generated by executing the ETL notebook on the original `hotel_bookings.csv` source file, recorded as a formal execution log and audit trail.

```
ETL PIPELINE - HOTEL BOOKING DEMAND DATASET

[STEP 1] EXTRACTING DATA...
  Rows loaded      : 119,390
  Columns loaded   : 32
  Memory usage     : 29.15 MB

[STEP 2] DATA QUALITY ANALYSIS...
  Missing Values Summary:
    - children: 4 missing (0.0%)
    - country: 488 missing (0.41%)
    - agent: 16,340 missing (13.69%)
    - company: 112,593 missing (94.31%)
  Duplicate rows detected : 31,994
  ADR range: [-6.38, 5400.00]
  ADR negative values     : 1
  Zero-guest records      : 180
  Overall cancel rate     : 37.04%

[STEP 3] DATA CLEANING...
  After drop_duplicates      : 87,396 rows
  Missing country filled with mode : PRT
  After removing zero-guest records : 87,230 rows
  After removing negative ADR      : 87,229 rows
  After removing ADR outliers (>335.00): 87,144
  Total rows removed : 32,246 (27.01%)

[STEP 4] FEATURE ENGINEERING...
  Features engineered: total_nights, total_guests,
    estimated_revenue, revenue_lost, arrival_date,
    season, lead_time_category, room_mismatch,
    has_children, risk_score, risk_category

[STEP 5] FINAL VALIDATION...
  Final shape      : (87144, 44)
  Remaining nulls   : 0
  Cancellation rate : 27.52%
  City Hotel records : 53,263
  Resort Hotel records : 33,881
  Date range        : 2015-07-01 to 2017-08-31
  Est. total revenue : EUR 34,312,805
  Revenue lost (cancel) : EUR 11,427,753
  Risk Category Distribution:
    Low Risk      237
    Medium Risk   38,147
    High Risk     36,517
    Very High Risk 12,243

[STEP 6] LOADING - EXPORTING TO EXCEL...
  Saved to : hotel_bookings_cleaned.xlsx
  File size : ~14,000 KB
=====
ETL PIPELINE COMPLETED SUCCESSFULLY
=====
```

4.10 Project Repository

All materials related to this section are publicly available in the project's GitHub repository. The repository contains the following resources:

- **ETL Notebook:** the complete Google Colab Jupyter Notebook (.ipynb) containing the full ETL pipeline documented in this section, ready to execute on any raw copy of the Hotel Booking Demand dataset.
- **Original Dataset:** the raw hotel_bookings.csv file as sourced from Kaggle, provided for full reproducibility of results.
- **Cleaned Dataset:** the output file hotel_bookings_cleaned.xlsx generated by the pipeline which serves as the direct data source for the Power BI dashboard in Section 5.

Repository: <https://github.com/juangalvez7077/Hotel-Bookings->

4.11 Section Summary and Handoff to Section 5

The ETL pipeline documented in this section takes the raw Hotel Booking Demand dataset - 119,390 records across 32 variables - and produces a clean, analytically enriched Excel file of 87,144 records and 44 variables, with zero remaining quality issues. The 32,246 records removed (27.01%) are all accounted for and justified, with the large majority coming from PMS-generated duplicate records rather than genuine data losses.

Beyond cleaning, the pipeline adds eleven engineered features that are specifically designed to power the Power BI dashboard in Section 5. These are not generic additions - each one was built with a specific visualisation or analytical need in mind. Without them, the dashboard would be limited to the 32 original variables, which on their own do not support risk scoring, revenue quantification, or seasonal analysis in a format that Power BI can use directly.

The output file hotel_bookings_cleaned.xlsx is the single data source for everything in Section 5. All KPI cards, trend charts, risk scoring visuals, and revenue analysis in the dashboard pull from this file. This ensures that there is a clear, traceable line from the raw PMS export all the way through to the final business intelligence output.

5. ANALYTICAL MODEL AND DASHBOARD

5.1 Overview

The main deliverable of this project is a four-page interactive Power BI dashboard built on the cleaned and enriched dataset produced by the ETL pipeline in Section 4. Rather than presenting static reports, the dashboard turns raw reservation data into something a revenue manager can actually act on - without needing any technical background. Each page is designed around a specific business question, and the answers come directly from the data.

Everything in the dashboard pulls from a single source: hotel_bookings_cleaned.xlsx, which contains 87,144 validated reservation records across 44 variables, including the 11 features engineered in the ETL pipeline. All KPI cards, trend charts, and risk visuals are calculated dynamically using DAX measures defined within the Power BI data model.

5.2 Page 1 - Executive Overview

Page 1 is designed for general management. The question it answers is simple: how healthy is our booking portfolio right now?

At the top of the page, five KPI cards give an immediate read on the most important numbers: 87,144 total reservations, a cancellation rate of 27.52%, an average daily rate of EUR 106, EUR 11.39M in revenue lost to cancellations, and an average lead time of 80 days. More than one in four bookings is being cancelled, and the financial impact is significant enough to justify a serious management response.

Below the KPIs, a dual-axis line chart tracks monthly reservation volumes and cancellation rates from July 2015 to August 2017. Both are trending upward, which means the cancellation problem is getting worse, not stabilising. The summer peaks in 2016 and 2017 are clearly visible and match what you would expect from a leisure-heavy Portuguese hotel market.

Two bar charts sit side by side comparing the City Hotel and the Resort Hotel. The City Hotel has more reservations (53,263 vs 33,881) but also cancels more (30.10% vs 23.46%). These are meaningfully different profiles that probably need different management strategies rather than a one-size-fits-all approach.

On the right, a stacked column chart shows the top 10 countries by reservation volume broken down by hotel type. Portugal dominates, followed by the UK, France, Spain, and Germany. This is useful for any marketing team trying to figure out which origin markets to prioritise or which ones to be more cautious about.

5.3 Page 2 - Cancellation Deep Dive

Page 2 goes deeper into the drivers of cancellation. It is aimed at revenue management teams and tries to answer the question: which types of booking are actually causing the problem?

Starting with market segment - Online TA has a cancellation rate approaching 40%, well above the portfolio average of 27.52%. Corporate and Direct bookings are below 15%. The channel a booking comes through turns out to be one of the strongest predictors of whether it will cancel, which makes sense given that OTAs actively promote free cancellation as a feature.

Deposit type tells a similar story. Non Refund bookings show a rate close to 100%, but this is a known anomaly in the dataset - these appear to be administrative records rather than real guest cancellations and should not be taken at face value. No Deposit bookings, which make up the vast majority of the portfolio, cancel at around 40%. Refundable bookings cancel much less. Some financial commitment at booking time does seem to make a difference.

By customer type, Transient customers - the largest group - cancel the most. Contract and Group bookings are far more reliable. Again, the most commercially accessible segments are also the most volatile, which creates a real tension for acquisition strategy.

Lead time is probably the clearest pattern in the whole dashboard. Cancellation rates go up consistently the further in advance a booking is made - from around 8% for bookings made less than a week before arrival to over 40% for bookings made more than six months out. This has direct practical implications for deposit policy and re-confirmation campaigns.

At the bottom of the page, a matrix heat map crosses market segment against lead time category to show cancellation rates for every combination. Online TA bookings with very long lead times hit 52% - the highest in the entire dataset. Direct and Corporate stay below 26% even at the longest lead times. This kind of granularity is what allows targeted interventions rather than blanket policies that affect everyone equally.

5.4 Page 3 - Revenue and Seasonality

Page 3 puts a financial number on everything identified in Page 2. It is built for commercial and finance teams and answers the question: when does the cancellation problem cost us the most money?

The monthly revenue trend shows strong seasonality - a clear peak above EUR 2.5M in August 2016 and a sustained upward trend through 2017. As the hotel grows, the absolute cost of cancellations grows with it. This is not a stable problem; it gets more expensive every year if left unaddressed.

The revenue lost column chart makes this concrete. In peak summer months, revenue lost to cancellations exceeds EUR 1M. The seasonal pattern mirrors total revenue almost exactly, which means cancellation risk is highest precisely when it hurts the most commercially. Getting ahead of this before the high season starts is not optional - it is the highest-value window for any intervention.

On the ADR comparison, the City Hotel averages EUR 110 versus EUR 99 for the Resort Hotel. This small difference helps explain why the City Hotel accounts for a disproportionate share of total revenue lost even though the two properties have a similar number of cancellations in absolute terms.

Summer accounts for nearly 30,000 reservations - almost double the volume of Winter. With demand this concentrated in one season, even a modest improvement in cancellation management during peak months would have a significant impact on full-year revenue.

5.5 Page 4 - Risk Scoring and Recommendations

Page 4 is where the dashboard moves from analysis to action. It uses the `risk_score` and `risk_category` features built in the ETL pipeline to answer the question: which specific bookings should we prioritise, and what should we do about them?

Looking at the distribution, most bookings fall into Medium Risk (38,147) and High Risk (36,517). Very High Risk accounts for 12,243 reservations, while only 237 are classified as Low Risk. The scoring framework is deliberately conservative - it flags risk rather than assuming bookings are safe by default, which makes sense given the scale of the cancellation problem.

High Risk bookings account for nearly EUR 6M in revenue lost, and Very High Risk adds another EUR 3M on top. Together that is over EUR 9M of the total EUR 11.39M - which means that targeting just these two categories would cover the vast majority of the recoverable revenue opportunity.

The risk score is a domain knowledge-based framework designed to identify which bookings generate the most revenue loss - and on that basis it performs well, with High and Very High Risk bookings accounting for over EUR 9M of the total EUR 11.39M lost. Like any composite scoring model, the weights assigned to each factor reflect empirical patterns in the data rather than a single definitive formula, and the relationship between score and observed cancellation rate is not perfectly linear across all categories. This is an inherent characteristic of domain-knowledge proxies and does not affect the practical utility of the tool for revenue management prioritisation.

The five operational recommendations on this page come directly from the data:

First, deposit policy reform: require a deposit for Online TA bookings with a lead time of more than three months. This combination produces the highest cancellation rates in the portfolio, and a deposit creates a financial commitment that reduces cancellation propensity.

Second, re-confirmation campaigns: contact High and Very High Risk bookings around 30 days before arrival. This gives revenue managers enough time to identify likely cancellations and re-sell the capacity before it is too late.

Third, overbooking calibration: adjust the overbooking buffer based on the monthly risk distribution of the forward booking portfolio rather than a fixed historical average. A uniform buffer systematically underestimates risk in summer when high-risk bookings are concentrated.

Fourth, direct channel development: invest in incentives to shift bookings from OTA channels to direct. Direct bookings cancel at below 15% compared to 40% for Online TA. Even at a lower ADR, the improvement in reliability generates a better revenue outcome on a risk-adjusted basis.

Fifth, risk score recalibration: in a next phase, replace the domain-knowledge-based weights with statistically derived feature importances from a logistic regression or gradient boosting model. This would improve the discriminative power of the score and allow individual-level probability estimates rather than broad risk categories.

5.6 UX and Design Decisions

From the start, the dashboard was built with non-technical users in mind. Revenue managers are not data scientists, and the design reflects that.

Every chart title uses plain language rather than technical field names. Cancellation Rate by Lead Time is immediately understandable; `is_canceled` by `lead_time_category` is not. This might seem like a small thing, but it makes a real difference to whether a tool actually gets used.

The navy blue background and consistent colour palette across all four pages keeps the visual experience coherent. Lighter blue tones are used for breakdowns and comparisons. Moving between pages does not feel disorienting because the visual language stays the same throughout.

Pages 2, 3, and 4 all use a 2x2 layout that groups related charts together. Reservation volume sits next to cancellation rate. Revenue trend sits next to revenue lost. This pairing means users can see two sides of the same story without having to scroll or switch views.

Cross-filtering works across all visuals on each page. Clicking on a bar in the market segment chart automatically filters every other chart on the page to show only that segment. This kind of interactivity means a revenue manager can answer follow-up questions on the spot without asking anyone for help.

Slicers at the top of every page let users filter by hotel type, arrival year, market segment, and country of origin simultaneously. A manager who wants to look specifically at UK Online TA bookings in 2017 can get there in a few clicks, and every chart on the page updates instantly.

Finally, the four-page structure maps to the four main audiences identified in Section 1 - general management, revenue management, commercial and finance, and operations. Each group lands on the page most relevant to them first, without having to wade through content that was designed for someone else.